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Technical Report – Audio Power Amplifier

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1. Functional Blocks of the Amplifier

- **Input Network & AC Coupling:**

The input network conditions the incoming audio signal by blocking any DC components and defining the amplifier's input impedance. This stage ensures that only the AC signal is applied to the active circuitry, preventing external DC offsets from affecting the internal biasing conditions and overall stability.

- **Differential Amplifier:**

The differential input stage compares the input signal with the feedback signal derived from the amplifier output. This stage performs three essential tasks: voltage amplification of the small audio signal, generation of the correct phase for the global feedback loop, and common-mode noise rejection. By amplifying only the difference between the two signals, the differential stage contributes to stable operation, improved linearity, and reduced distortion.

- **Voltage Amplification Stage:**

After the differential pair, the signal is applied to the Voltage Amplification Stage, which provides most of the amplifier's voltage gain. A small capacitor (C4) connected to this stage forms a Miller compensation network that improves high-frequency stability, defines the dominant pole, and prevents oscillation when driving reactive loads such as loudspeakers. This ensures stable operation across the audio frequency range.

- **Driver Transistors**

Before the output stage, the audio signal passes through driver transistors (BC550/BC560). This stage provides sufficient current gain to drive the power transistors and isolates the small-signal stages from the high current demands of the loudspeaker.

- **Output Stage: Darlington Power Transistors**

The output stage uses complementary Darlington power transistors to deliver high current and power directly to the loudspeaker. The high current gain of the Darlington devices allows the amplifier to deliver approximately 60 W without overloading the driver stage. Proper heatsinking is critical due to power dissipation.

- **Global Negative Feedback Network**



A resistive feedback network connects the amplifier output to the differential input stage. This global negative feedback stabilises gain, reduces distortion, controls bandwidth, and minimises DC offset, ensuring linear and stable operation.

- **Power Supply Requirements**

The amplifier requires a dual-rail $\pm 24V$ power supply capable of delivering several amperes. Adequate filtering and star grounding are necessary to prevent hum or oscillation. Strong heatsinking, proper transistor isolation, and quick-blow fuses are recommended for safe and reliable operation.

2. Role of the Differential Input Stage

The differential input stage compares the input signal with the feedback signal from the amplifier output. It amplifies only the difference between these two signals, providing common-mode noise rejection and generating the correct phase relationship for feedback operation. This stage contributes to stable amplifier operation and low distortion.

3. Voltage Amplification Stage and Current Amplification Stages

The voltage amplification stage is the Voltage Amplification Stage (VAS), located after the differential input stage. It provides most of the amplifier's voltage gain.

The current amplification stages are the driver stage and the output stage, which supply the high current required to drive the loudspeaker load.

4. Stability Provided by the Voltage Amplification Stage and Miller Compensation

A small capacitor connected around the VAS forms a Miller compensation network. This network improves high-frequency stability, defines the dominant pole of the amplifier, and prevents oscillations when driving reactive loads such as loudspeakers.

5. Current and Voltage Behaviour of the Darlington Output Stage

The Darlington output stage provides very high current gain with near-unity voltage gain. The output voltage follows the signal from the driver stage, while the Darlington transistors supply the large current required by the loudspeaker. Complementary NPN and PNP devices conduct during opposite signal half-cycles.

6. Operating Class of the Amplifier



The amplifier uses a complementary push-pull output stage with NPN and PNP Darlington transistors. This type of output stage is commonly used in audio amplifiers and typically operates in Class AB, reducing crossover distortion while maintaining acceptable efficiency.

7. Power Supply, Thermal, and Layout Considerations

The amplifier requires a dual-rail ± 24 V power supply capable of delivering several amperes. Strong heatsinking, proper transistor isolation, and the use of quick-blow fuses are recommended for thermal safety. Good filtering, star grounding, and careful PCB layout are essential to avoid hum and oscillation.